PyGameMaker — Tutorial 15: Fruit Fusion

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Fruit Fusion: Catch and Merge!** (6 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

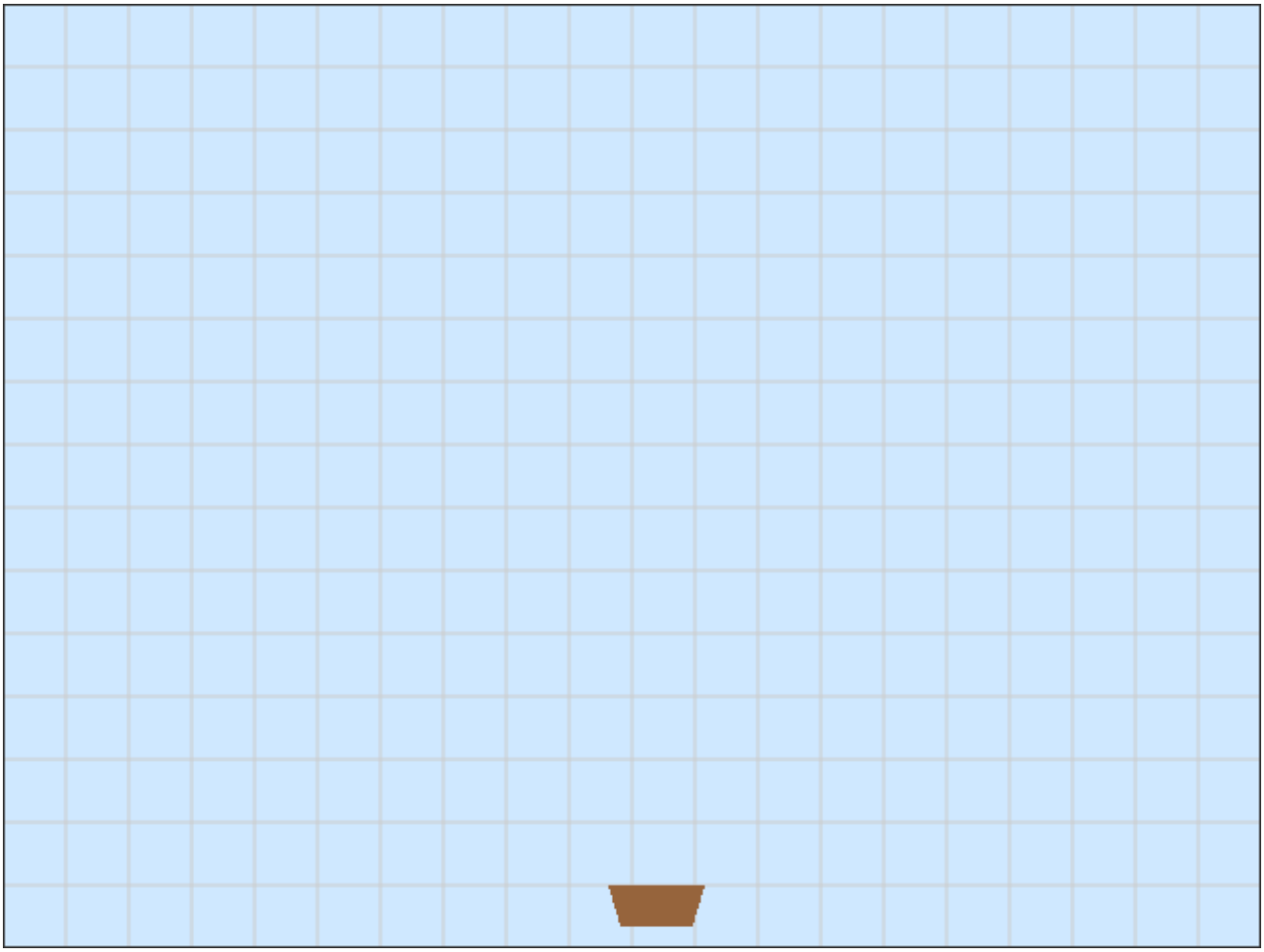
What You Will Make

A basket that always holds one fruit, starting with a cherry. Cherries, strawberries and oranges fall from the sky. Catch a fruit that matches what you are holding and it fuses into the next, bigger fruit. Fuse all the way up to a watermelon and you win!

Phase 1: Moving Basket

- [ ] **1.** Make a new project FruitFusion.
- [ ] **2.** Create a sprite spr_basket_empty (56x40, a brown bowl shape) and an object obj_player with that sprite.
- [ ] **3.** Rename the project's starting room (room0) to room_main, set its **width to 640** and **height to 480** in its properties, and place obj_player near the bottom center.
- [ ] **4.** In obj_player: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -5*; **Right Arrow (held)** with *5*; **Keyboard: No key** with *Stop movement*.

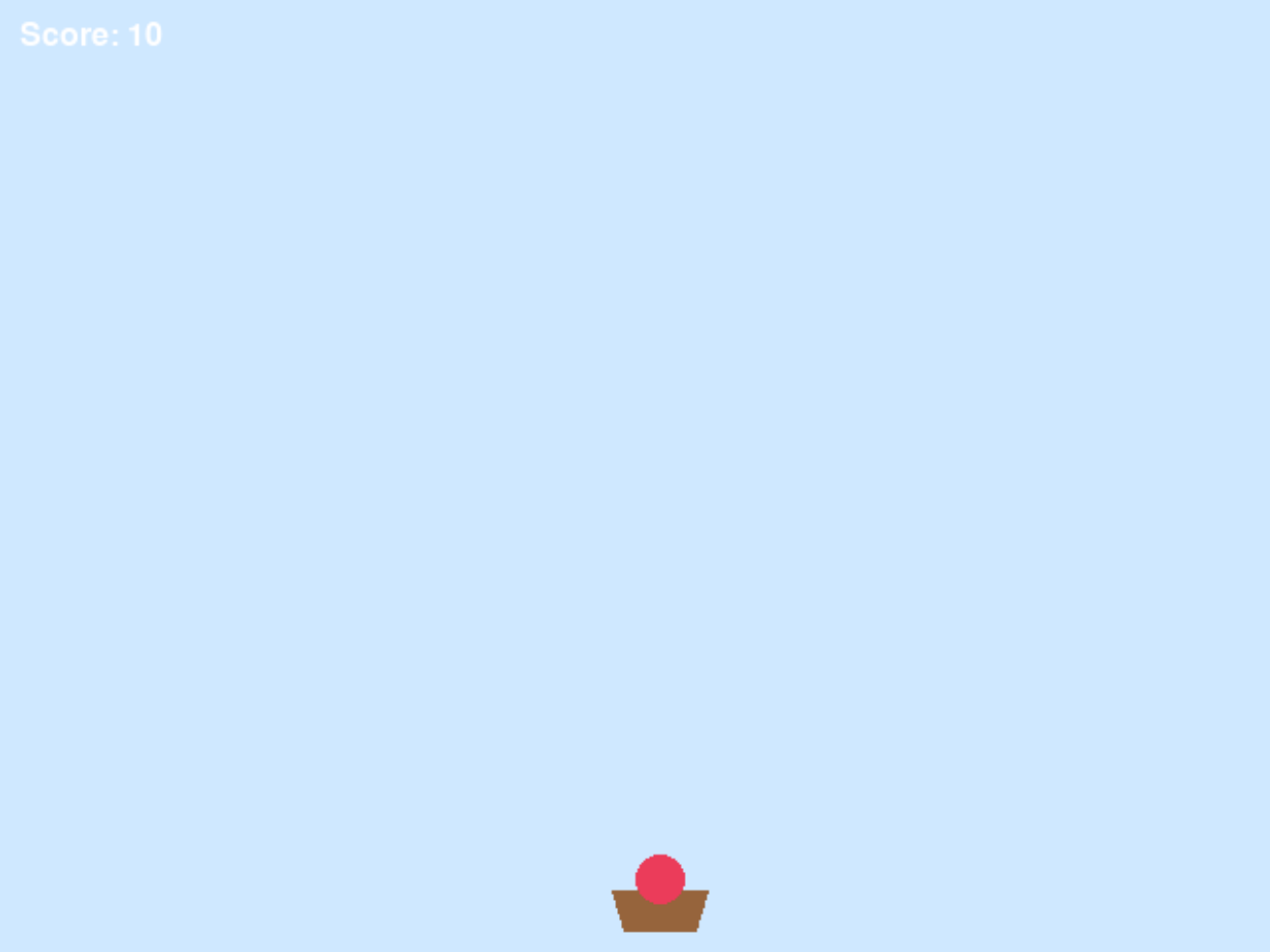
You should see: press **F5**. Left and Right move the basket, and it stops when you let go.



Phase 2: First Fusion

- [ ] **5.** Create a sprite spr_fruit_cherry (20x20, a red circle) and an object obj_fruit_cherry with a **Create** event (*Set vertical speed 3*) and an **Outside Room** event (*Destroy this instance*). **Outside Room** is not its own block — in the Events category, drag out **Other event:** and pick "Outside Room" from its dropdown.
- [ ] **6.** Create obj_spawn_cherry (no sprite) with a **Create** event (*Set alarm 0 to 90*) and an **Alarm 0** event (*Create instance of obj_fruit_cherry at x: random integer from 0 to 620, y: 0*, then *Set alarm 0 to 90* again). Place one in room_main. (620 = room width 640 minus the cherry's own width 20, so it never spawns hanging off the edge.)
- [ ] **7.** Change obj_player's sprite to a new sprite spr_basket_cherry (a bowl with a cherry in it) — the basket now starts out already holding a cherry.
- [ ] **8.** Before this step: **Set Variable** is not turned on by default. Go to Tools > Configure Action Blocks..., expand **Control**, check **Set Variable**, click OK. Now in obj_player: **Create** event sets *score to 0* and *variable held_level to 1*; **Draw** event does *Draw score at x 10, y 10*.
- [ ] **9.** Create a sprite spr_basket_strawberry (56x40, a bowl with a strawberry in it) — you'll need it in the next step.
- [ ] **10.** Add **Collision with obj_fruit_cherry**: *If held_level equals 1 then* [*set held_level to 2*, *add 10 to score*, *set sprite to spr_basket_strawberry*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

You should see: press **F5**. Catch a falling cherry — the basket turns into a strawberry and the score jumps by 10.

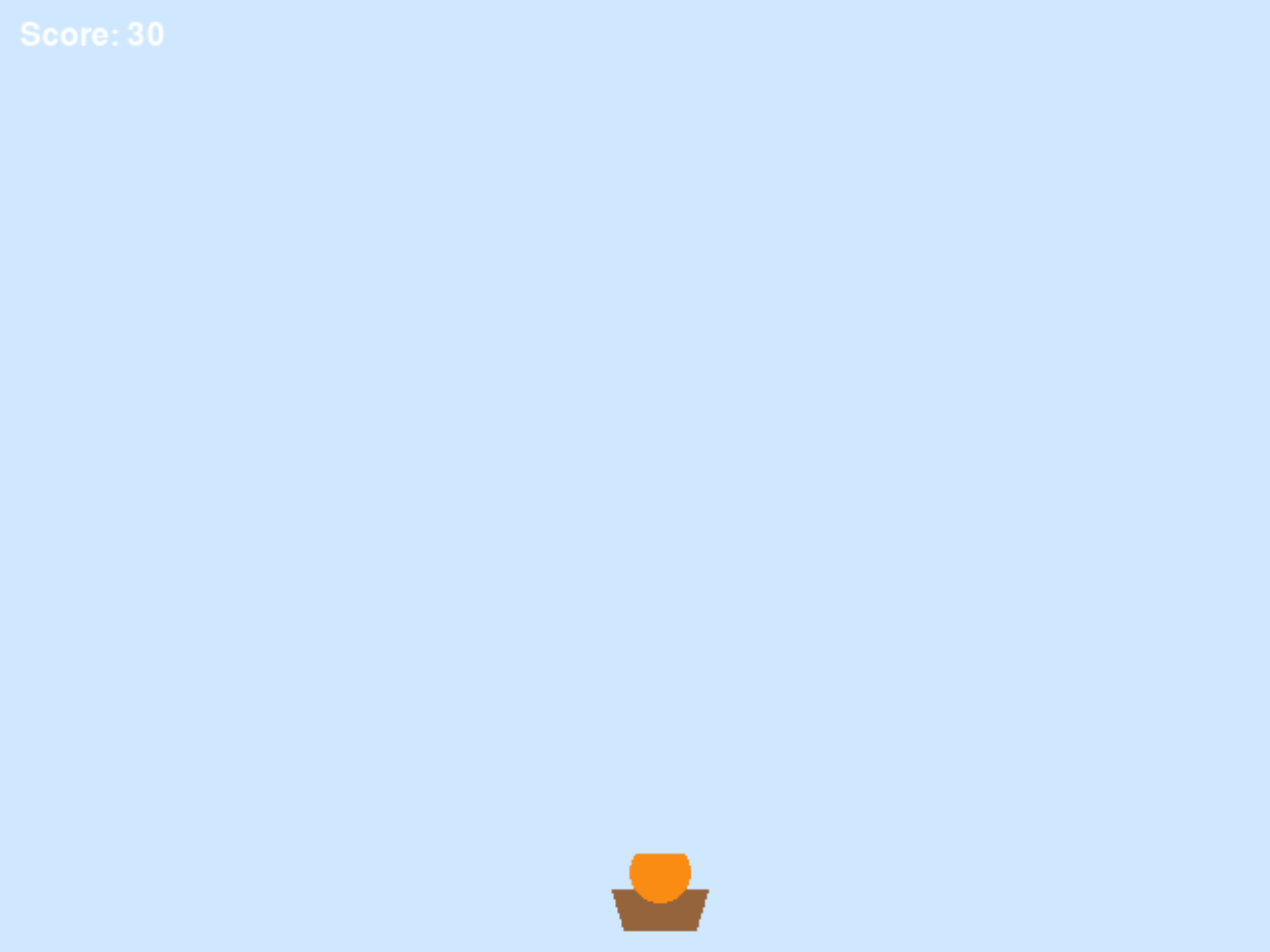


Phase 3: Second Fusion

This phase repeats the exact same recipe as Phase 2, with "cherry" changed to "strawberry" and the numbers changed from 1/2 to 2/3.

- [ ] **11.** Create a sprite spr_fruit_strawberry (26x26, a pink-red circle) and an object obj_fruit_strawberry, built exactly like obj_fruit_cherry.
- [ ] **12.** Create obj_spawn_strawberry (alarm 0 to 120 this time; random x from 0 to 614, since the strawberry is 26 wide) and place one in room_main.
- [ ] **13.** Create a sprite spr_basket_orange (56x40, a bowl with an orange in it) — you'll need it in the next step.
- [ ] **14.** Add **Collision with obj_fruit_strawberry** on obj_player: *If held_level equals 2 then* [*set held_level to 3*, *add 20 to score*, *set sprite to spr_basket_orange*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

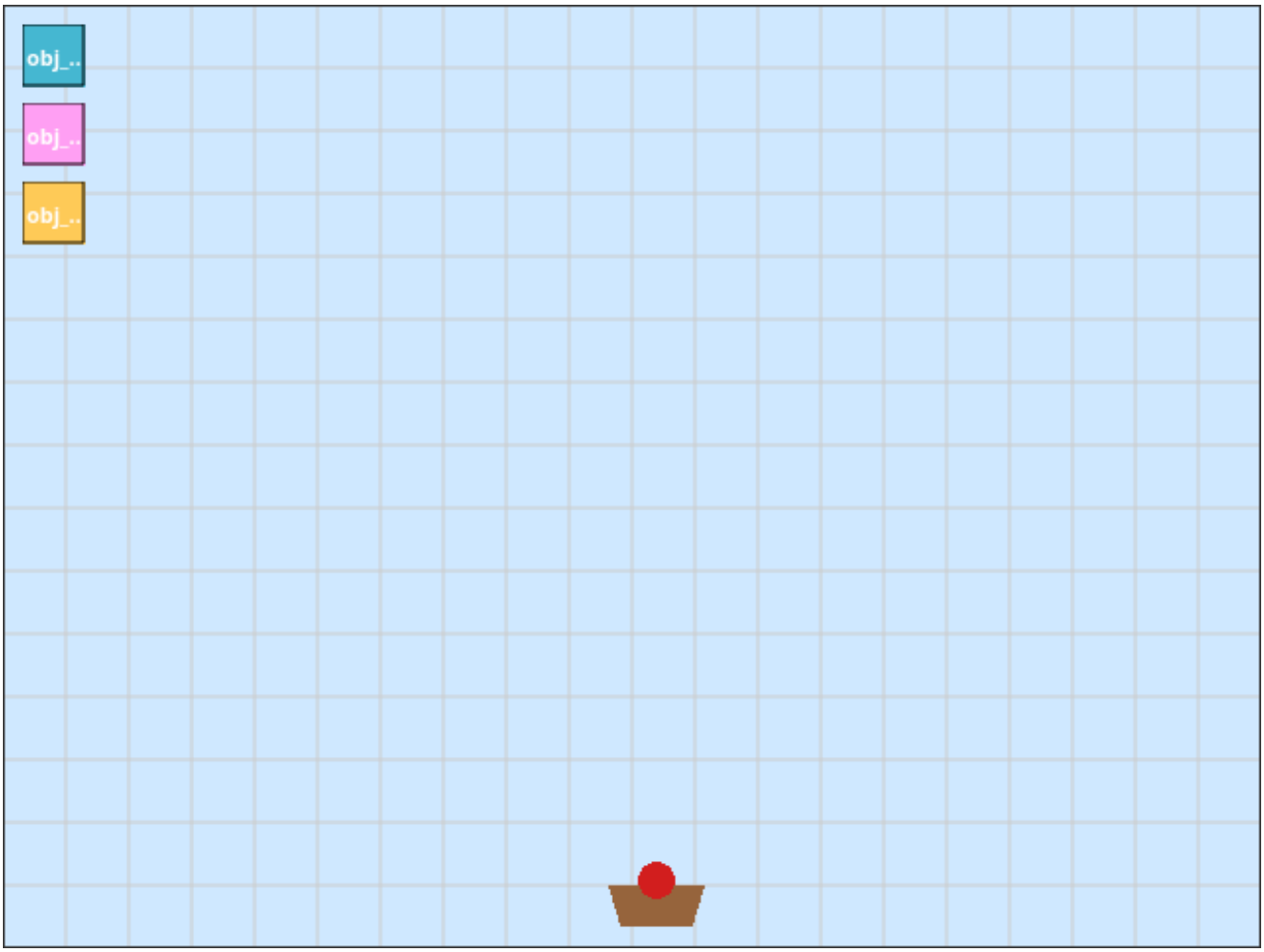
You should see: press **F5**. Catch a cherry, then a strawberry — the basket ends up an orange, having scored 30 points.



Phase 4: Winning

- [ ] **15.** Create a sprite spr_fruit_orange (32x32, an orange circle) and an object obj_fruit_orange, built exactly like the others. There is no falling watermelon sprite or spawner — a watermelon is only ever something your basket becomes.
- [ ] **16.** Create obj_spawn_orange (alarm 0 to 150; random x from 0 to 608, since the orange is 32 wide) and place one in room_main.
- [ ] **17.** Create obj_win_text. **Draw** event: *Draw text* "YOU WIN! You made a watermelon!" at x 120, y 220 and "Press SPACE to play again" at x 190, y 260. **Key press: space**: *Restart game*.
- [ ] **18.** Create room_win with a bright background color, and place one obj_win_text instance in it.
- [ ] **19.** Create a sprite spr_basket_watermelon (56x40, a bowl with a green, watermelon-striped circle in it) — you'll need it in the next step.
- [ ] **20.** Add **Collision with obj_fruit_orange** on obj_player: *If held_level equals 3 then* [*set held_level to 4*, *add 50 to score*, *set sprite to spr_basket_watermelon*, *Go to room room_win*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

> **You should see:** press **F5**. Chain all three fusions and the YOU WIN! room appears. SPACE starts a new game.



Bonus (optional): Import Ready-Made Art

Page 6 of the tutorial shows how to swap any of your drawings for ready-made pictures instead, if you'd rather not draw. Your game works exactly the same either way — this is only worth doing once everything above already works.

Stuck?

What is wrong	Most likely cause	What to do
"Set variable" is not in the Control category	It is not turned on by default	Tools > Configure Action Blocks..., expand **Control**, check **Set Variable**, click OK
The basket never changes picture	set_sprite is missing from the *If* branch, or it points at the wrong sprite name	Check the exact sprite name in *Set sprite to*
Catching a matching fruit only gives 1 point	held_level is not actually equal to the number the *If* checks, or the condition uses the wrong operator	Check *If held_level equals* uses the right number for that phase (1, then 2, then 3)
The basket is stuck as a cherry forever	The merge's *Otherwise* branch accidentally changed held_level too	Only the *If* branch should touch held_level and the sprite

Fruit piles up and never disappears	The fruit's **Outside Room** event is missing	Add *Destroy this instance* in Outside Room
No fruit ever falls	The spawner object has no instance placed in room_main, or its alarm was never set in Create	Place one spawner instance; check *Set alarm 0* is in **Create**, not somewhere else
Catching an orange does not win	The *Go to room room_win* action is in the wrong place, or the room name is misspelled	It must be inside the *If held_level equals 3* branch, spelled exactly room_win
The win screen is blank	obj_win_text was not placed in room_win, or its Draw event is missing	Place one instance; check the Draw event exists
SPACE does nothing on the win screen	The win room has no object with a **Key press: space** event	Add the event to obj_win_text

Challenges

- **Try this (5 minutes):** change the fruit fall speed, or how often a spawner fires.
- **Push further:** add a fourth spawner and fruit type between orange and watermelon.
- **Invent:** add lives that go down when you mismatch too many times in a row.

Vocabulary

Term	What it means
Held fruit	The one fruit the basket is currently carrying, remembered in held_level
Fusion	Catching a matching fruit so it grows into the next, bigger fruit
If / Else	A block that does one thing when something is true, and a different thing when it is not
Spawner	An invisible object whose only job is to create other objects on a timer
Outside Room	An event that fires when an instance leaves the room — used here to clean up missed fruit

Check Yourself

Write your answers in the notes below.

1. Why does the basket need a variable like held_level at all?
2. What happens if you catch a cherry while already holding an orange?
3. Why is there no sprite or spawner for a falling watermelon?

My Notes